Siyologics Commerce Platform

A Full-Stack Direct-to-Customer Operations System

Built and designed as a scalable prototype platform for a product-based business to manage product showcasing, direct customer orders, online payments, shipment tracking, inventory operations, and admin workflows through a unified web experience.

FLASK | PYTHON | MYSQL | HTML5 · CSS3 · JS | FLASK-MAIL | RAZORPAY

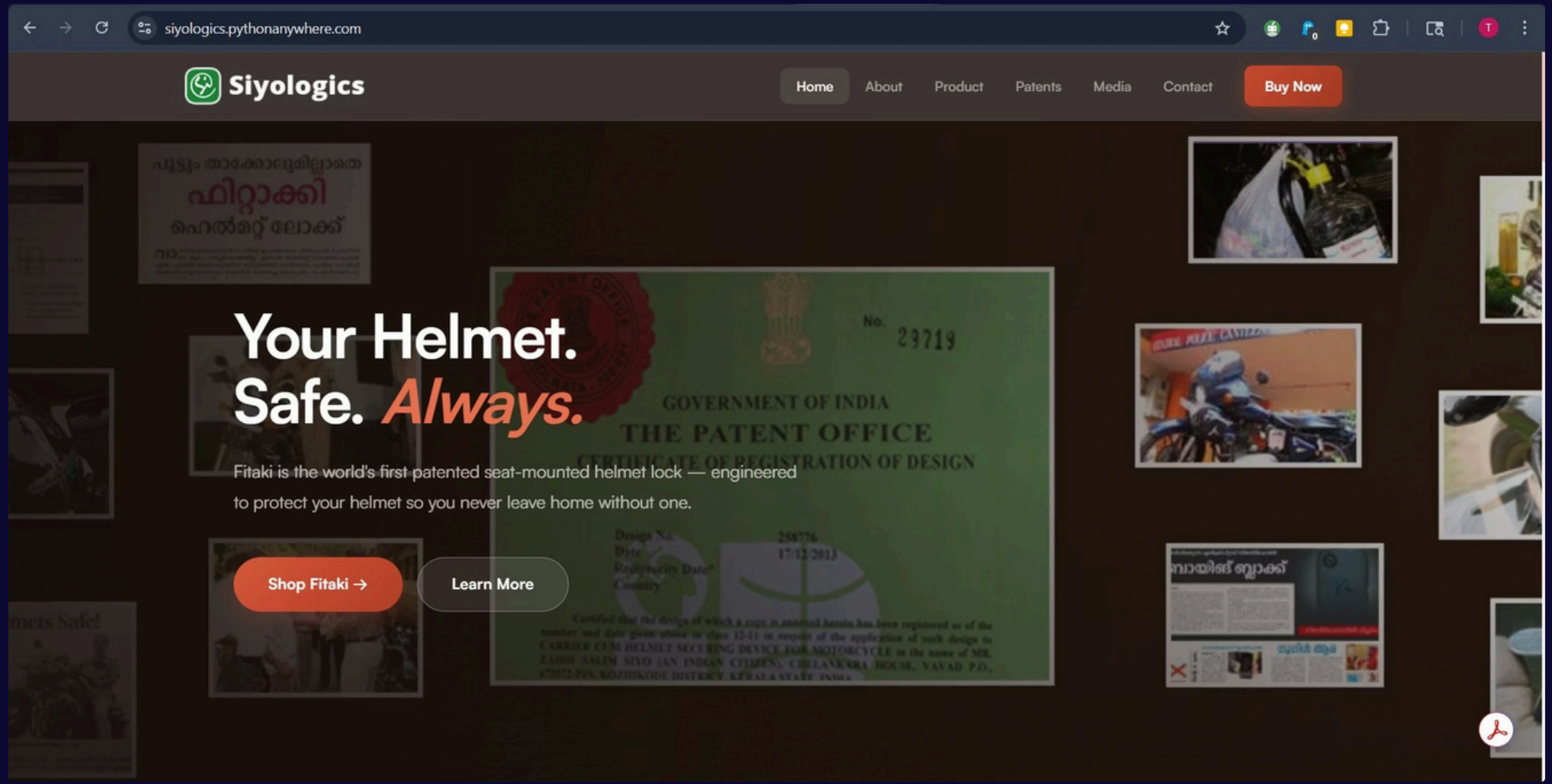
Developed By

Thaman Siyo

Full Stack Web Developer

Live Prototype

https://siyologics.pythonanywhere.com/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Project Overview

SiyologicsCommerce Platform was developedasa direct-to-customer operational system for managing product visibility, online ordering, payments, shipment tracking, and internalbusinessworkflowsthrough asinglecentralizedplatform. The project combines customer-facing commerce functionality with backend operational management, creating a connected workflow between ordering, payments, inventory, and shipping operations.

Business Problem

The business previously relied heavily on external marketplaces and manual operational workflows, creating multiple limitations:

- No centralized order management
- No direct customer ownership
- Limited branding control
- Manual shipment handling
- No inventory workflow integration
- Lack of operational visibility

The absence of an independent platform also restricted long-term scalability and customer engagement.

Solution

The platform was designed to handle both customer interaction and backend business operations through a unified full-stack system.

1

Direct Ordering

Product ordering with dynamic pricing

2

Payment Integration

Razorpay secure checkout

3

Shipment Tracking

Real-time order status updates

4

Email Automation

Automated transactional notifications

5

Admin Dashboard

Inventory & order lifecycle management

Customer Features & User Flow

Core Customer Features

Product Showcase

Exploreproductdetails,pricing, features, and usage information directly through the platform.

Secure Payments

Validated ordersprocessedthrough Razorpay Checkout with dynamic server-side pricing validation.

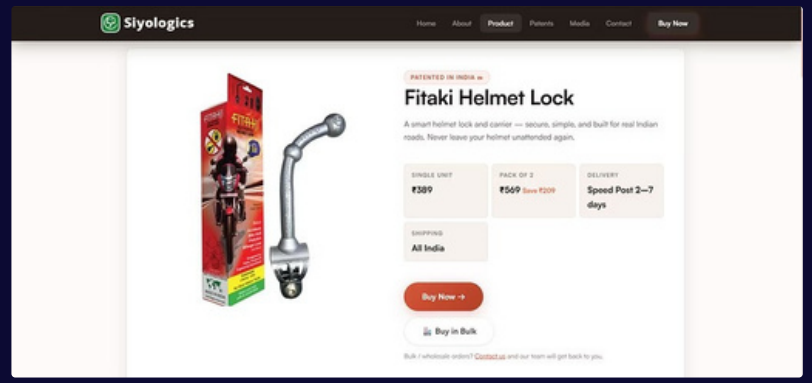
Shipment Tracking

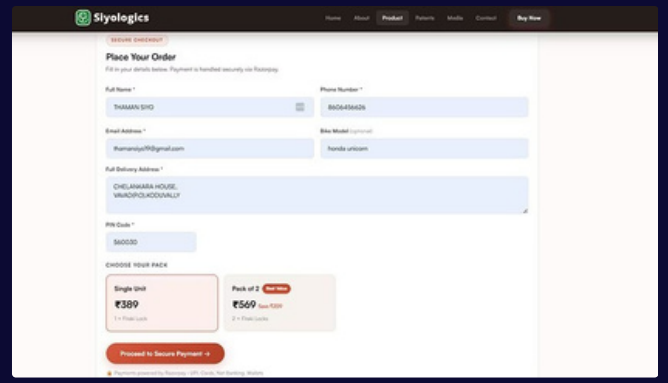
TrackorderprogressusingOrder ID and Email — no account login required.

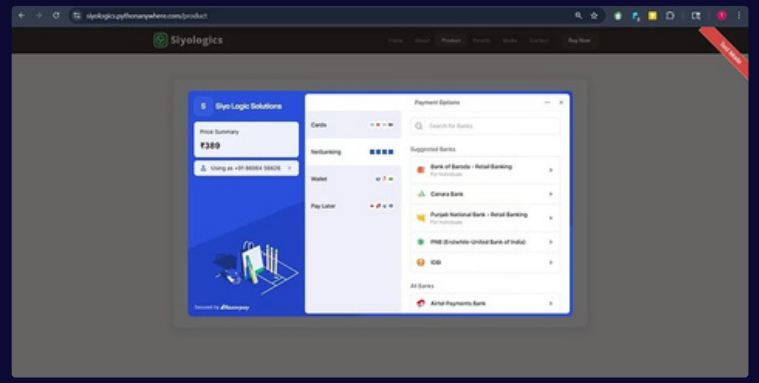
User Flow

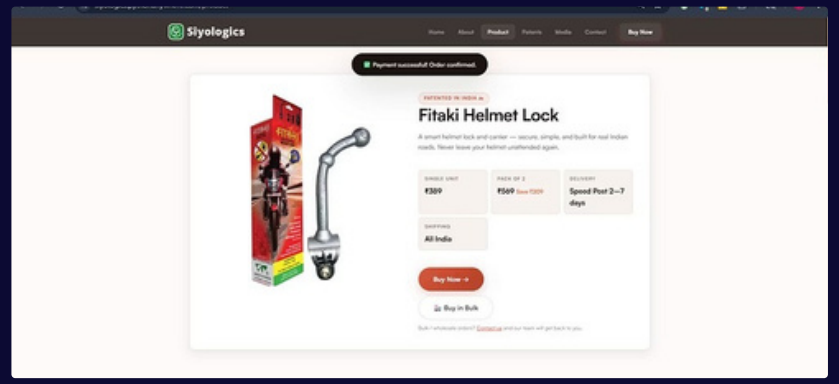
A minimal-friction checkout experience with integrated payment processing and real-time operational synchronization from browse to delivery.

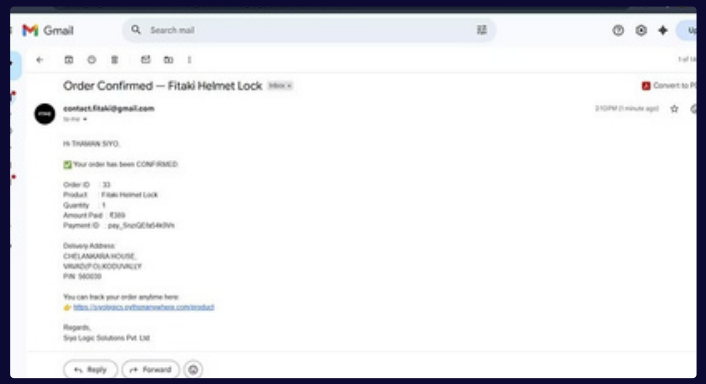
User Flow - Screenshots

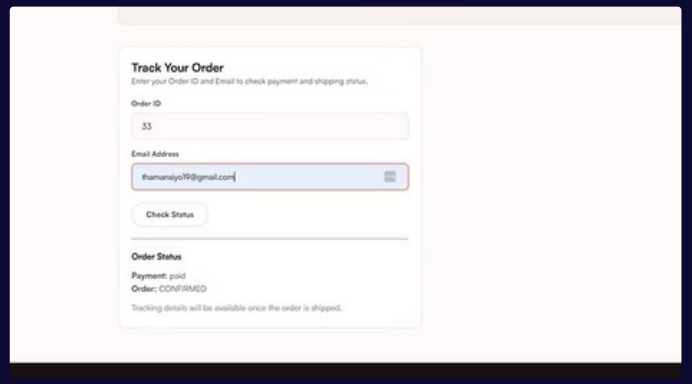












Admin Operations & Dashboard

Theadmin systemwasbuilttomanage the completeorder lifecycle frompayment confirmationtofinal delivery, with backend workflow integration connecting payment verification, inventorysynchronization, order management, shipment handling,and customercommunication.

The platform validates operational transitions and prevents invalid shipment updates for unpaid orders.

KPI Monitoring

- Total paid orders
- Revenue summaries
- Recent transactions
- Inventory availability

Stock Management

- Auto-update after payments
- Manual stock additions
- Inventory level monitoring
- Multi-product architecture

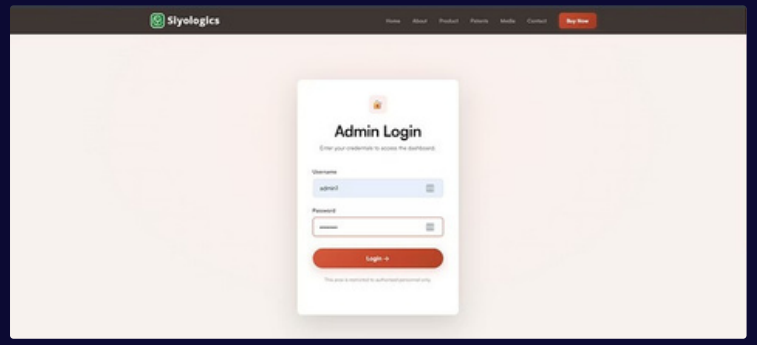
Shipment Operations

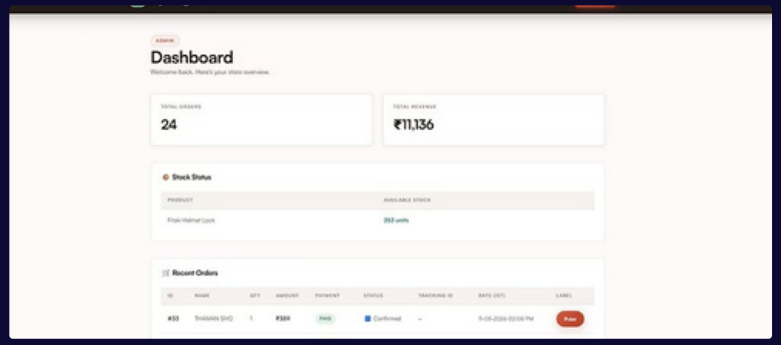
- Assign tracking IDs
- Update shipment status
- Manage courier details
- Print shipping labels

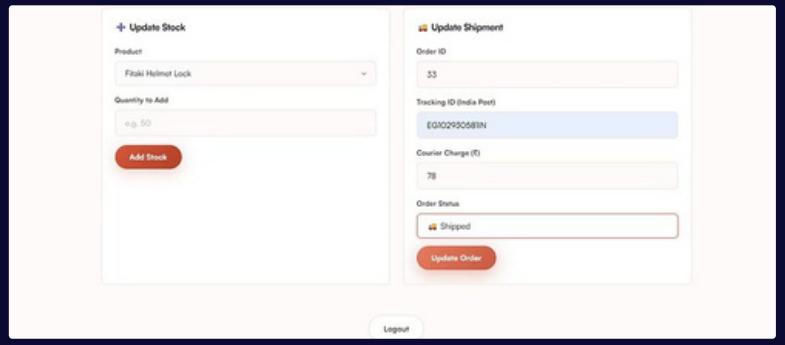
Automated Notifications

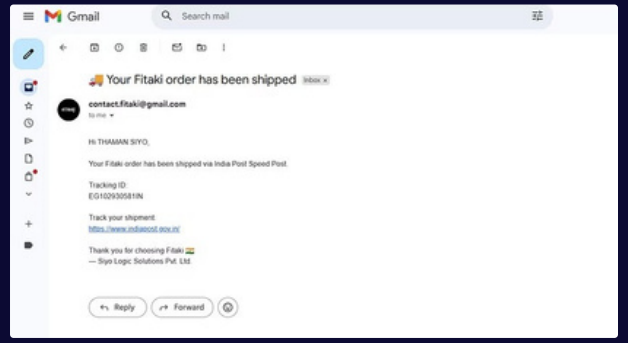
- Dispatch email alerts
- Order confirmations
- Shipment status updates
- Admin operational alerts

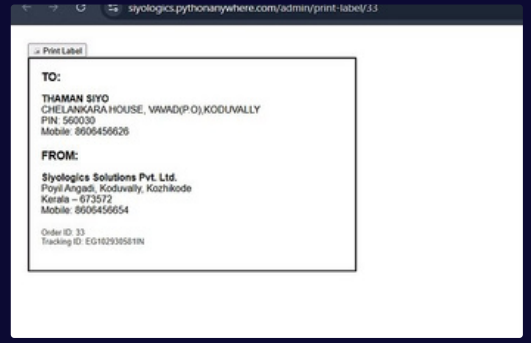
Admin Operations-Screenshot

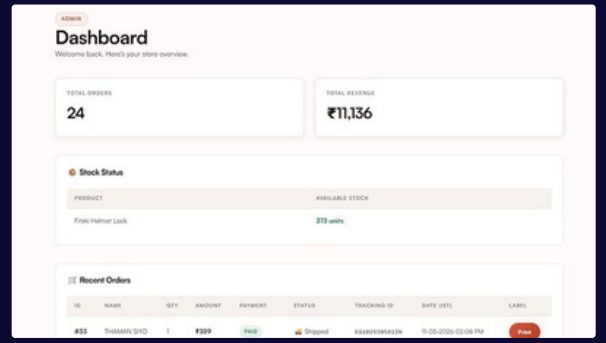


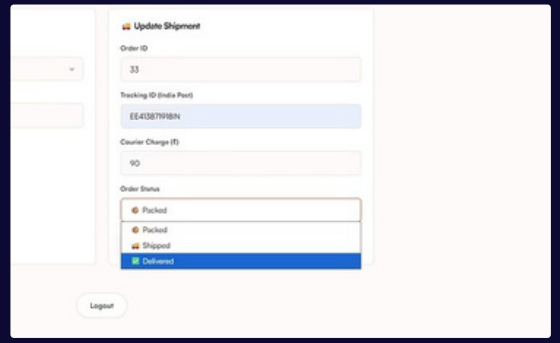












Tech Stack & System Architecture

Core Technology Stack

Frontend
HTML5 · CSS3 · JavaScript · Bootstrap · Jinja2 Templates

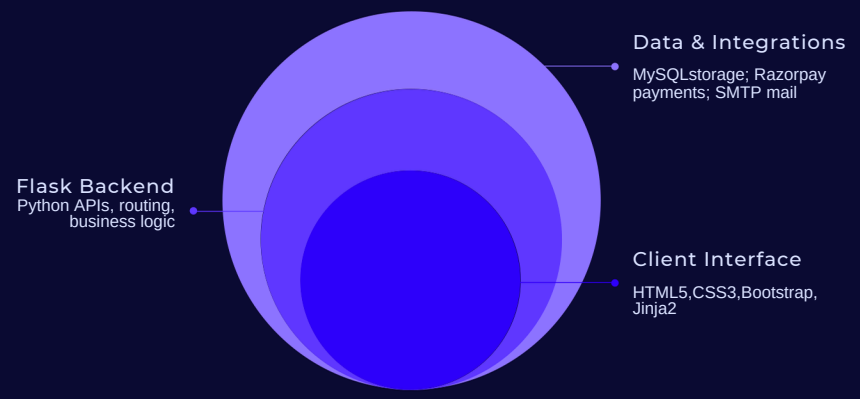
Backend
Flask(Python) · mysql-connector-python

Database
MySQL—relational data management

Integrations
Razorpay · Flask-Mail · Gmail SMTP

Deployment
PythonAnywhere · Git · GitHub

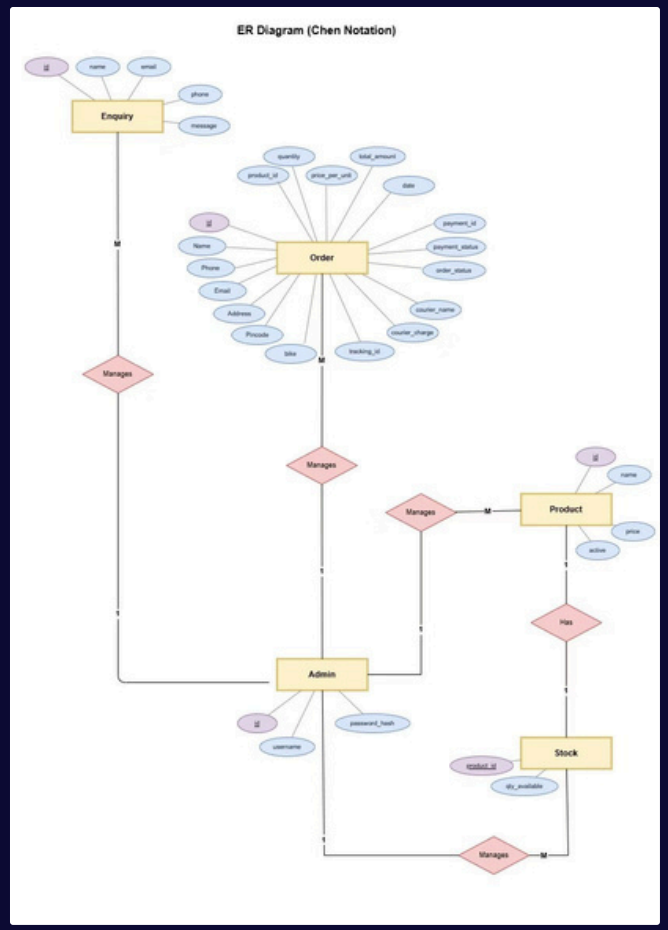
System Architecture Flow



The platform follows a layered full-stack architecture connecting frontend interfaces, backend business logic, database operations, and external integrations into a unified operational workflow. The Flask backend handles route management, session authentication, order processing, payment handling, inventory synchronization, shipment workflows, email automation, and database communication.

Database Design & Entity Relationships

Theplatformuses arelationalMySQLdatabasearchitecturetomanagecustomeroperations,inventoryworkflows,payment handling, shipment tracking, and administrative controls through connectedentity relationships.



Core Entities

Orders

Customer information, payment status, shipment lifecycle, tracking details, and pricing records.

Products & Stock

Product details, SKU configuration, listing status, and inventory quantity management per product.

Enquiry

Customer contact and enquiry submissions for operational communication.

Admins

Secure admin authentication and operational access control management.

Structured relational mapping · Inventory synchronization · Parameterised queries · Expandable multi-product architecture

Payments, Security & Engineering Challenges

Payment Workflow

The platform integrates Razorpay Checkout for handling secure online transactions during the order process. Successful payments trigger order confirmation, payment verification, stock synchronization, and automated email workflows. Failed transactions are handled separately through dedicated backendvalidation logic.

Security Implementation

Credential Protection

Sensitive credentialsmanaged via environment variables — no hardcoded values.

Secure Authentication

Admin authenticationuses hashed password verification through Werkzeug security utilities.

Database Security

Alldatabaseoperationsuse parameterized SQL queries to reduce injection vulnerabilities.

Session Protection

Adminroutes areprotectedusing session-based authentication checks.

Engineering Challenges

01

Payment-State Synchronization

Ensuringinventory updatesonly trigger after successful payment confirmation — not before.

02

Duplicate Callback Handling

Preventing repeated payment callbacks from creating duplicate order updates in the system.

03

Shipment Lifecycle Validation

Restricting invalidorder status transitions for unpaid or incomplete orders.

04

Inventory Consistency

Maintaining synchronized stock updates during successful order processing workflows.

The platform was designed to maintain transactional consistency across payments, inventory, and shipment handling throughout the full operational lifecycle.

UI Showcase & Platform Interface

The platform interface was designed to maintain a clean operational workflow across both customer-facing pages and backend administrative controls. The UI focuses on structured navigation, operational clarity, responsive layouts, streamlined interaction flow, and simplified order management.

Customer Interface

- Homepage & product showcase
- Integrated order form
- Razorpay payment flow
- Shipment tracking interface
- Contact & enquiry workflow

Admin Interface

- KPI dashboard
- Order management table
- Stock control system
- Shipment update workflow
- Tracking & dispatch management

Structured Navigation

Clear page hierarchy and intuitive navigation paths across all customer and admin interfaces.

Responsive Design

Mobile-friendly layouts built with Bootstrap ensuring consistent experience across all devices.

Workflow Efficiency

Minimal-friction interaction design prioritizing usability and operational speed for both users and admins.

Future Scope & Project Summary

Future Enhancements

01

Multi-Product Commerce

Expandcatalogsupportbeyondsingle-product architecture.

02

Customer Account System

User registration, login, and order history management.

03

Analytics & Reporting

Advanceddashboardwithoperational insights and inventory forecasting.

04

Mobile Application

Nativemobileintegrationwithreal-time shipment tracking APIs.

05

Cloud Infrastructure

Scalableclouddeploymentwith multi-admin operational roles.

Project Summary

Siyologics CommercePlatform demonstrates the development of a full-stack operational commerce system connecting customer workflows, payment processing, inventory management, shipment handling, and backend administration into a unified platform architecture.

The project combines practical business workflows with real-world engineering concepts including payment integration, transactional operations, database design, authentication systems, and operational automation.

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